

Dyadic Multiresolution and Sampling Frontiers for the Fabius–Rvachev System

**Exact rational wavelets, odd-lattice reconstruction, binomial layers,
Bessel expansions, and accelerated sinc energies**

Repository-aware analytic and computational investigation

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Abstract

The bounded Fabius distribution function F , Rvachev’s compactly supported function up , and the entire sinc product that is the Fourier transform of up possess unusually many mutually illuminating descriptions. This report first audits all 92 TeX paths recursively present under `Analysis/FabiusFunction/docs` at the pinned repository commit. Existing probability, convolution, functional–differential, Volterra, finite Thue–Morse spline, q -binomial, Fourier product, canonical-product, Fourier–Legendre, half-integer cosine, moment, Lambert– W , Bell-polynomial, and inverse-quantile representations are catalogued as prior corpus material, not reannounced as discoveries.

The main new theorem package is dyadic and multiresolution in nature. Exact rational values of F at dyadic points give a uniformly convergent Faber–Schauder expansion with rational coefficients, an exact-rational piecewise-affine limit for the inverse Fabius function, a Haar expansion of the density, and a locally uniformly convergent Haar–sinc² expansion of the infinite sinc product. Haar Parseval then gives a positive rational series for

$$A_2 = \int_0^1 F(x)^2 dx = \frac{1}{2\pi} \int_0^\infty \prod_{j \geq 1} \text{sinc}^2\left(\frac{t}{2^j}\right) dt,$$

together with an all-orders dyadic tail expansion. This turns an arithmetically opaque sinc integral into monotone exact rational approximants and produces a high-order acceleration yielding about sixty stable decimal digits from level 16 dyadic data.

Several additional representation families are derived. Expanding every uniform coordinate into fair binary digits reorganizes the Fabius random series into independent binomial layers and proves the product identity

$$\prod_{j \geq 1} \text{sinc}\left(\frac{t}{2^j}\right) = \prod_{n \geq 2} \cos\left(\frac{t}{2^n}\right)^{n-1}.$$

Paley–Wiener sampling reconstructs the same entire function solely from its odd π -lattice samples; the known even zero lattice then yields a hierarchy of new sample sum rules. Chebyshev and Gegenbauer coefficients become Bessel transforms of the sinc product. Dyadic Bernstein polynomials give exact-rational polynomial limits for F and up , while their Fourier images are finite sums of Kummer functions. Finally, finite approximants are identified with sublevel volumes and hyperplane sections of weighted cubes. Proofs, status labels, conjectures, a Lean formalization roadmap, and a fully commented Python verification program are included.

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1 Audit boundary, status discipline, and contribution map

1.1 Pinned recursive corpus

The source boundary of this report is the Git commit

f7daf632f2db839fe091de0d080e22a9b50c286a

of Vladimir Reshetnikov’s ProveIt repository [16]. The audit root is Analysis/FabiusFunction/docs. A previously reconstructed pinned Git manifest contained 87 recursive TeX paths. During the audit the branch advanced by 25 commits; pinning the new head and comparing trees revealed five added TeX paths and no removals. Thus the current boundary contains exactly 92 TeX paths. The complete list, the five additions, and the modified-path ledger are included as corpus_manifest_current.txt in the accompanying archive.

The five additions are three large integral/transform manuscripts and two generated numerical fragments. They materially change the novelty boundary: finite and complex-power antiderivatives, fractional Volterra calculus, quantile beta transforms, Cauchy–Stieltjes hierarchies, inverse-Fabius resolvents, autocorrelations, order statistics, and a rational Jacobi continued fraction were already present by the pinned commit. Those results are therefore baseline here.

The audit did not treat archived revisions as independent corroboration. Historical copies were read for mathematically distinctive material, then deduplicated by lineage. Imported TeX transcriptions of the papers of Arias de Reyna and of work on lacunary products were included in the collision search. Targeted repository searches were also made for the exact phrases and formula shapes used below. No occurrence was found for the Faber–Schauder midpoint-defect expansion, the rational Haar energy series, the triangular independent-binomial decomposition, the odd-sample cardinal formula, the Chebyshev–Bessel sinc integral, the dyadic Bernstein–Kummer transform, or the weighted-cube section representation.

1.2 Meaning of the status labels

Three labels are used throughout.

- **Corpus baseline.** means that the result, perhaps in a different normalization, is explicit in the audited corpus or one of its imported sources.
- **New deduction relative to the pinned corpus.** means that a proof is supplied here and that no explicit counterpart was located in the pinned corpus after formula-level searches. It is not a claim of absolute publication priority.
- **Conjecture.** means that no proof is claimed. Exact computations and structural evidence are stated separately from the conjecture.

This distinction is especially important in the Fabius project: a new representation may be an elementary consequence of deep existing identities, while a visually similar formula may fail because it suppresses the nonconstant periodic phase in the endpoint asymptotics.

1.3 Existing material deliberately excluded from novelty claims

The following are treated as established inputs.

Layer	Audited prior material
Basic functions	Separation of the bounded CDF F , the compactly supported even bump up, and the signed global extension $\mathcal{F}$; uniqueness, smoothness, support, symmetry, strict shape, and flat endpoints.
Probability	Weighted sums of independent uniform coordinates, finite atomic and spline approximants, weak convergence, exact support bounds, and reflection.
Arithmetic	Rational moments, reciprocal-power values $F(2^{-n})$, arbitrary dyadic evaluation, Thue–Morse formulas, q -Pochhammer and Gaussian-binomial forms, denominator and 2-adic information.
Fourier analysis	Infinite sinc and cosine products, Fourier inversion, Poisson and half-integer sampling identities in physical space, integer zeros with multiplicity $1 + \nu_2(n)$, canonical products, sharp real-axis decay, spectral zeta and heat trace.
Approximation	Finite Thue–Morse splines, repeated integration, Fourier–Legendre expansion, exact Strang–Fix reproduction, positive quadrature, and Richardson acceleration of truncated products.
Integral transforms	Mellin, Laplace, Fourier, Cauchy, Stieltjes, logarithmic, correlation, spacing, order-statistic, inverse-quantile, and fractional-resolvent transforms.
Endpoint theory	Correct lower-Lambert coordinate, nonconstant periodic Gamma–zeta correction, all-order Bell-polynomial expansion, inverse-quantile reversion, and phase-aware Richardson extraction.

1.4 New theorem package

The report proves the following linked results.

1. An exact rational Faber–Schauder expansion of F , with sharp 4^{-J} interpolation scale, sign symmetry, and curvature recovery; inversion gives exact-rational piecewise-affine approximants to $G = F^{-1}$.
2. A rational Haar expansion of F' and up, and a corresponding Haar–sinc² series for the entire Fourier transform.
3. A positive rational series for $A_2 = \int_0^1 F^2$, a certified geometric tail bound, and a complete asymptotic expansion of its dyadic level energies and tails.
4. An independent-binomial layer representation obtained by regrouping the binary digits of the uniform coordinates, including a finite combinatorial CDF and probability polynomial.
5. An odd- π -lattice Shannon reconstruction of the sinc product and a hierarchy of sample identities forced by the higher even-zero multiplicities.
6. Chebyshev and Gegenbauer expansions whose coefficients are Bessel transforms of the sinc product; the Legendre specialization turns known rational coefficients into exact Bessel-product integrals.
7. Exact-rational dyadic Bernstein limits and finite Kummer-function formulas for their Fourier transforms.
8. Weighted-cube sublevel-volume and hyperplane-section limits unifying the probabilistic, spline, and Fourier-slice pictures.

2 Normalization and established representation atlas

2.1 The three functions

Let $F : \mathbb{R} \rightarrow [0, 1]$ denote the bounded Fabius distribution function [2, 4, 5]:

$$F(x) = 0 \quad (x \leq 0), \quad F(x) = 1 \quad (x \geq 1), \quad (1)$$

with

$$F'(x) = 2F(2x) \quad (0 < x < \tfrac{1}{2}), \quad F(x) + F(1 - x) = 1 \quad (0 \leq x \leq 1). \quad (2)$$

Rvachev's function is [3]

$$\text{up}(y) = \begin{cases} F(1 - |y|), & |y| \leq 1, \\ 0, & |y| > 1. \end{cases} \quad (3)$$

It is even, nonnegative, C^∞ , supported on $[-1, 1]$, and normalized by $\int \text{up} = 1$. The conversion formulas are

$$F'(x) = 2 \text{up}(2x - 1), \quad F(x) = \text{up}(x - 1) \quad (0 \leq x \leq 1), \quad (4)$$

and, for $0 \leq x \leq 1$,

$$\text{up}(x) = 1 - F(x) = F(1 - x). \quad (5)$$

The signed global continuation $\mathcal{F}$ satisfies $\mathcal{F}'(x) = 2\mathcal{F}(2x)$ on all of $\mathbb{R}$ and agrees with F on $[0, 1]$. It is indispensable for global derivative masks, but the new multiresolution formulas below concern the bounded F and compactly supported up . We never silently substitute one global convention for another.

2.2 Probability and infinite convolution

Corpus baseline. Let $U_1, U_2, \dots$ be independent uniform random variables on $[0, 1]$ [1] and let $V_j = 2U_j - 1$, uniform on $[-1, 1]$. Then

$$X = \sum_{j=1}^{\infty} \frac{U_j}{2^j}, \quad Y = 2X - 1 = \sum_{j=1}^{\infty} \frac{V_j}{2^j}, \quad (6)$$

$F(x) = \mathbb{P}(X \leq x)$, and Y has density up . Consequently

$$\text{up} = \sum_{j=1}^{\infty} (2^{j-1} \mathbf{1}_{[-2^{-j}, 2^{-j}]}) \quad (7)$$

in the sense of weak limits and, in fact, in the smooth-density topology available from the corpus.

The survival interpretation

$$\text{up}(t) = \mathbb{P}(X > t), \quad 0 \leq t \leq 1, \quad (8)$$

immediately gives, for absolutely continuous H ,

$$\int_0^1 H'(t) \text{up}(t) dt = \mathbb{E}[H(X) - H(0)], \quad (9)$$

$$\int_0^1 H'(t) F(t) dt = H(1) - \mathbb{E}H(X). \quad (10)$$

This is the compact master identity behind many moment, Laplace, logarithmic, and resolvent integrals in the audited transform reports.

2.3 Functional–differential and Volterra forms

Corpus baseline. Integrating (2) gives the primitive self-similarity

$$F(x/2) = \int_0^x F(t) dt, \quad 0 \leq x \leq 1. \quad (11)$$

Repeated integration produces exact primitive ladders. The compact bump satisfies the global functional–differential equation

$$\text{up}'(x) = 2 \text{up}(2x + 1) - 2 \text{up}(2x - 1), \quad (12)$$

and all higher derivatives are finite translates with Thue–Morse signs. The current corpus extends these identities to integer and fractional Volterra orders and to broad arbitrary-kernel transmutation formulas.

2.4 Fourier, Laplace, product, and canonical-product forms

We use the angular Fourier convention

$$\Phi(t) = \int_{-1}^1 \text{up}(y) e^{-ity} dy. \quad (13)$$

The cycles-per-unit transform in the repository is $\widehat{\text{up}}(\xi) = \Phi(2\pi\xi)$. Throughout, $\text{sinc } z = \sin z/z$ with $\text{sinc } 0 = 1$.

Corpus baseline. Independence in (6) gives

$$\Phi(t) = \prod_{j=1}^{\infty} \text{sinc}\left(\frac{t}{2^j}\right). \quad (14)$$

The moment-generating function of X is therefore

$$M_X(s) = \mathbb{E}e^{sX} = \prod_{j=1}^{\infty} \frac{e^{s/2^j} - 1}{s/2^j} = e^{s/2} \Phi(is/2). \quad (15)$$

Its logarithm yields Bernoulli cumulants and Bell-polynomial moments.

The complete complex zero divisor is the nonzero lattice $2\pi\mathbb{Z}$, with multiplicity $1 + \nu_2(n)$ at $2\pi n$. Equivalently,

$$\Phi(t) = \prod_{n=1}^{\infty} \left(1 - \frac{t^2}{4\pi^2 n^2}\right)^{1+\nu_2(n)}, \quad (16)$$

where the symmetric genus-zero product converges normally. The corpus contains the associated spectral zeta, determinant, heat trace, zero jets, and sharp real-axis decay theory.

Fourier inversion gives

$$\text{up}(y) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \Phi(t) e^{ity} dt, \quad (17)$$

and symmetry gives the CDF representation

$$F(x) = \frac{1}{2} + \frac{1}{\pi} \int_0^{\infty} \Phi(t) \frac{\sin((2x-1)t)}{t} dt, \quad 0 < x < 1. \quad (18)$$

The decay is faster than every power, so these are ordinary convergent integrals rather than distributional shorthand.

The period-two extension of up has only the constant and odd Fourier modes:

$$\text{up}(y) = \frac{1}{2} + \sum_{m=0}^{\infty} \Phi((2m+1)\pi) \cos((2m+1)\pi y), \quad |y| \leq 1. \quad (19)$$

This physical-space cosine series is established corpus material. The frequency-space odd-sample reconstruction in [section 7](#) is a distinct, dual statement.

2.5 Finite splines, exact dyadic arithmetic, and limits

Corpus baseline. Put $X_N = \sum_{j=1}^N 2^{-j} U_j$ and let F_N be its CDF. Inclusion–exclusion for a weighted box gives the finite Thue–Morse spline

$$F_N(x) = \frac{2^{N(N+1)/2}}{N!} \sum_{k=0}^{2^N-1} (-1)^{\text{wt}_2(k)} \left(x - \frac{k}{2^N}\right)_+^N. \quad (20)$$

Because $0 \leq X - X_N \leq 2^{-N}$,

$$F_N(x - 2^{-N}) \leq F(x) \leq F_N(x). \quad (21)$$

This is simultaneously a probability limit, a piecewise-polynomial limit, and a finite alternating truncated-power representation.

Every dyadic value is rational. The reciprocal-power values $V_n = F(2^{-n})$ obey

$$V_0 = 1, \quad V_n = \frac{1}{2^{\binom{n}{2}} (2^n - 1)} \sum_{k=0}^{n-1} \frac{2^{\binom{k}{2}} V_k}{(n - k + 1)!}. \quad (22)$$

The repository further supplies finite Thue–Morse sums, q -Pochhammer and Gaussian-binomial forms, an executable highest-bit recursion for arbitrary $F(a/2^e)$, and exact denominator and valuation theorems. Those exact values will become the coefficients of the new Faber and Haar representations.

2.6 Moment, orthogonal-polynomial, and endpoint forms

Corpus baseline. The even moments

$$\mu_{2r} = \int_{-1}^1 y^{2r} \text{up}(y) dy \quad (23)$$

are rational and satisfy

$$\Phi(t) = \sum_{r=0}^{\infty} (-1)^r \mu_{2r} \frac{t^{2r}}{(2r)!}. \quad (24)$$

The even cumulants are

$$\kappa_{2r} = \frac{2^{2r-1} B_{2r}}{r(2^{2r} - 1)}, \quad \kappa_{2r+1} = 0, \quad (25)$$

and complete Bell polynomials convert the cumulants into the moments. The corpus also proves an absolutely and uniformly convergent Fourier–Legendre expansion of up with exact rational coefficients.

At the flat endpoints, the natural variable is the exact lower-Lambert coordinate [\[15\]](#)

$$\lambda(x) = -\frac{1}{\log 2} W_{-1}(-x \log 2), \quad \lambda 2^{-\lambda} = x. \quad (26)$$

The full endpoint expansion of F contains a nonconstant one-periodic Gamma–zeta term in this coordinate, and its coefficients are generated by Bell-polynomial algebra. The inverse function $G = F^{-1}$ inherits a phase-aware Lambert reversion. These subtle asymptotics are baseline; below we formulate only new multiresolution questions that retain, rather than average away, this periodic phase.

2.7 A compact dictionary of established representations

For later comparison, [table 2](#) records the principal established forms in one place.

Table 2: Established representation atlas used as input.

Object	Representation	Natural use
F	CDF of $\sum 2^{-j} U_j$	Monotonicity, coupling, quantiles, stochastic integrals.
up	Density of $\sum 2^{-j} V_j$	Symmetry, convolution, support, probability.
F	$F(x/2) = \int_0^x F$	Primitive ladders and fractional Volterra calculus.
up	Infinite convolution of uniform boxes	Smoothness and finite spline approximants.
F_N	Finite Thue–Morse truncated-power sum	Exact computation and certified limits.
$F(a/2^e)$	Rational recurrence, bit recursion, q -binomial sums	Arithmetic and exact dyadic evaluation.
Φ	Infinite sinc product	Global Fourier analysis and decay.
Φ	Canonical product over $2\pi\mathbb{Z}$ with multiplicities	Zero spectrum, determinants, zeta and heat trace.
up	Fourier inversion integral	Arbitrary real arguments and analytic estimates.
F	Sine-integral CDF inversion	Arbitrary arguments and endpoint cancellation.
up	Period-two odd cosine series	Physical-space spectral approximation.
up	Fourier–Legendre series	Orthogonal approximation with rational coefficients.
M_X	Infinite exponential quotient product	Moments, cumulants, Laplace asymptotics.
F, G	Mellin, Stieltjes, beta-quantile, spacing transforms	Integral calculus and inverse-function identities.
F, G	Lower-Lambert periodic transseries	Sharp endpoint and inverse-endpoint analysis.

3 Exact rational Faber–Schauder representation

3.1 Dyadic tents and midpoint defects

Following the classical Haar–Faber–Schauder multiresolution lineage [6, 7, 8, 14], for $m \geq 0$ and $0 \leq k < 2^m$, put

$$\ell_{m,k} = \frac{k}{2^m}, \quad r_{m,k} = \frac{k+1}{2^m}, \quad \xi_{m,k} = \frac{2k+1}{2^{m+1}}, \quad h_m = 2^{-m-1}. \quad (27)$$

The normalized tent on this cell is

$$\Lambda_{m,k}(x) = \max(1 - 2^{m+1}|x - \xi_{m,k}|, 0). \quad (28)$$

Its support is $[\ell_{m,k}, r_{m,k}]$, its height is one, and its area is h_m . Define the midpoint defect

$$c_{m,k} = F(\xi_{m,k}) - \frac{F(\ell_{m,k}) + F(r_{m,k})}{2}. \quad (29)$$

Every argument in (29) is dyadic, so

$$c_{m,k} \in \mathbb{Q}. \quad (30)$$

Theorem 3.1 (Exact rational Faber–Schauder series). *For $J \geq 0$, let*

$$S_J F(x) = x + \sum_{m=0}^{J-1} \sum_{k=0}^{2^m-1} c_{m,k} \Lambda_{m,k}(x). \quad (31)$$

Then $S_J F$ is exactly the piecewise-linear interpolant of F on the dyadic mesh $2^{-J}\mathbb{Z} \cap [0, 1]$. Consequently

$$F(x) = x + \sum_{m=0}^{\infty} \sum_{k=0}^{2^m-1} c_{m,k} \Lambda_{m,k}(x) \quad (32)$$

uniformly on $[0, 1]$.

Proof. The level-zero affine interpolant through $(0, F(0)) = (0, 0)$ and $(1, F(1)) = (1, 1)$ is x . Passing from the mesh of level m to that of level $m + 1$ leaves all old nodes fixed and introduces one midpoint in each cell. The unique correction supported on that cell, zero at the endpoints and having the required midpoint value, is precisely $c_{m,k} \Lambda_{m,k}$. Induction proves the interpolation statement. Uniform convergence follows from uniform continuity of F , since the error of linear interpolation on a mesh of diameter 2^{-J} is bounded by twice the modulus of continuity at that scale. The stronger quantitative bound in [theorem 3.6](#) follows from $F \in C^2$. $\square$

New deduction relative to the pinned corpus. The point of [theorem 3.1](#) is not the abstract existence of a Schauder expansion; it is that every coefficient is an explicitly computable rational number supplied by the repository’s exact dyadic arithmetic. The partial sums are therefore exact rational piecewise-linear functions, not floating-point projections.

Corollary 3.2 (Rvachev form). *For $|y| \leq 1$,*

$$\text{up}(y) = 1 - |y| + \sum_{m=0}^{\infty} \sum_{k=0}^{2^m-1} c_{m,k} \Lambda_{m,k}(1 - |y|). \quad (33)$$

The series is uniformly convergent on $[-1, 1]$ and is zero outside that interval after the usual compact-support extension.

Proof. Substitute $x = 1 - |y|$ in (32) and use (3). $\square$

3.2 Coefficient kernel, symmetry, signs, and curvature

Proposition 3.3 (Triangular-kernel formula). *The midpoint defect has the exact integral representation*

$$c_{m,k} = -\frac{1}{2} \int_{-h_m}^{h_m} (h_m - |u|) F''(\xi_{m,k} + u) \, du. \quad (34)$$

In particular,

$$|c_{m,k}| \leq 2^{-2m}. \quad (35)$$

Proof. The second symmetric difference satisfies

$$F(\xi + h) + F(\xi - h) - 2F(\xi) = \int_{-h}^h (h - |u|) F''(\xi + u) du.$$

Equation (34) follows from the definition of $c_{m,k}$. Moreover, $\|F''\|_\infty = 8$: from $F'(x) = 2 \uparrow(2x - 1)$ one has $\|F'\|_\infty = 2$, and on the left half $F''(x) = 4F'(2x)$, with the reflected formula on the right half. Since the triangular kernel has integral h_m^2 , we obtain $|c_{m,k}| \leq 4h_m^2 = 2^{-2m}$. $\square$

Proposition 3.4 (Reflection and sign pattern). *For every m, k ,*

$$c_{m,2^m-1-k} = -c_{m,k}. \quad (36)$$

Moreover, $c_{0,0} = 0$, and for $m \geq 1$,

$$c_{m,k} < 0 \quad (0 \leq k < 2^{m-1}), \quad c_{m,k} > 0 \quad (2^{m-1} \leq k < 2^m). \quad (37)$$

Proof. Reflection follows by applying $F(1 - x) = 1 - F(x)$ to the three nodes in (29). On $(0, 1/2)$, $F''(x) = 4F'(2x) > 0$, so F is strictly convex and every midpoint lies strictly below its chord. On $(1/2, 1)$ the reflected function is strictly concave. The level-zero defect vanishes because $F(1/2) = 1/2$. $\square$

Proposition 3.5 (Curvature recovery). *Uniformly in k as $m \rightarrow \infty$,*

$$-2^{2m+3} c_{m,k} = F''(\xi_{m,k}) + \mathcal{O}(4^{-m}). \quad (38)$$

More generally, for every fixed $Q \geq 0$,

$$c_{m,k} = - \sum_{r=0}^Q \frac{F^{(2r+2)}(\xi_{m,k}) h_m^{2r+2}}{(2r+2)!} + \mathcal{O}(h_m^{2Q+4}), \quad (39)$$

uniformly in k .

Proof. Insert the Taylor expansion of $F''(\xi + u)$ into (34). Odd powers integrate to zero and

$$\int_{-h}^h (h - |u|) u^{2r} du = \frac{2h^{2r+2}}{(2r+1)(2r+2)}.$$

The compact interval and smoothness of F make the remainder uniform. $\square$

The coefficients therefore encode exact dyadic samples of curvature. This is especially suggestive near the flat endpoints, where the repository’s Lambert–periodic asymptotics can be sampled through a purely rational sequence.

3.3 Sharp interpolation scale

Theorem 3.6 (Uniform error and sharp asymptotic constant). *For every $J \geq 0$,*

$$\|F - S_J F\|_\infty \leq 4^{-J}. \quad (40)$$

The constant is asymptotically sharp:

$$\boxed{\lim_{J \rightarrow \infty} 4^J \|F - S_J F\|_\infty = 1.} \quad (41)$$

Proof. On an interval of length $\Delta = 2^{-J}$, the classical linear-interpolation remainder gives

$$|F(x) - S_J F(x)| \leq \frac{\|F''\|_\infty}{8} \Delta^2 = \Delta^2 = 4^{-J}.$$

For the matching lower asymptotic, take the cell immediately to the right of $x = 1/4$ and evaluate at its midpoint. The interpolation remainder is $-F''(\eta_J)\Delta^2/8$ for some η_J in that cell. Since $\eta_J \rightarrow 1/4$ and $F''(1/4) = 8$, its absolute value divided by Δ^2 tends to one. $\square$

Remark 3.7. The estimate is stronger than a generic Faber–Schauder convergence theorem: it identifies the exact global interpolation scale and its limiting constant. The extremal curvature occurs at the dyadic points $1/4$ and $3/4$, so the sharp constant is itself compatible with exact dyadic arithmetic.

3.4 Low-level exact coefficients

The first nonzero rows illustrate both rationality and reflection:

$$(c_{1,k})_{k=0}^1 = \left(-\frac{13}{72}, \frac{13}{72}\right), \quad (42)$$

$$(c_{2,k})_{k=0}^3 = \left(-\frac{1}{32}, -\frac{1}{32}, \frac{1}{32}, \frac{1}{32}\right), \quad (43)$$

$$(c_{3,k})_{k=0}^7 = \frac{1}{2073600}(-3457, -28943, -28943, -3457, 3457, 28943, 28943, 3457). \quad (44)$$

At level four, the left half is

$$-\frac{1}{147456}(5, 149, 427, 571, 571, 427, 149, 5), \quad (45)$$

and the right half is its negative reversal. These rows were generated from (22) and the terminating dyadic bit recursion; no numerical integration entered.

3.5 A universal kernel dictionary

Uniform convergence permits termwise integration against every $K \in L^1(0, 1)$:

$$\int_0^1 K(x)F(x) dx = \int_0^1 xK(x) dx + \sum_{m=0}^{\infty} \sum_{k=0}^{2^m-1} c_{m,k} \int_0^1 K(x)\Lambda_{m,k}(x) dx. \quad (46)$$

When K belongs to a parameterized locally bounded family, the same identity is locally uniform in the parameter. Thus one exact coefficient table feeds polynomial moments, Laplace transforms, Fourier transforms, Stieltjes kernels, and fractional integral kernels.

For $p \in \mathbb{N}_0$, define

$$L_p(c, h) = 2h \sum_{r=0}^{\lfloor p/2 \rfloor} \binom{p}{2r} \frac{c^{p-2r} h^{2r}}{(2r+1)(2r+2)}. \quad (47)$$

This is exactly the moment of a tent:

$$\int_{c-h}^{c+h} x^p \left(1 - \frac{|x-c|}{h}\right) dx = L_p(c, h). \quad (48)$$

Hence

$$\int_0^1 x^p F(x) dx = \frac{1}{p+2} + \sum_{m=0}^{\infty} \sum_{k=0}^{2^m-1} c_{m,k} L_p(\xi_{m,k}, h_m). \quad (49)$$

Every summand is rational. This does not replace the more compact moment recurrences in the corpus; it gives a new positive-scale decomposition whose truncation error is controlled directly by (40).

Let

$$A(t) = \int_0^1 F(x) e^{-itx} dx. \quad (50)$$

The elementary tent transform is

$$\int_0^1 \Lambda_{m,k}(x) e^{-itx} dx = h_m e^{-it\xi_{m,k}} \operatorname{sinc}^2\left(\frac{th_m}{2}\right). \quad (51)$$

Therefore

$$A(t) = \frac{e^{-it}(1+it) - 1}{t^2} + \sum_{m=0}^{\infty} \sum_{k=0}^{2^m-1} c_{m,k} h_m e^{-it\xi_{m,k}} \operatorname{sinc}^2\left(\frac{th_m}{2}\right), \quad (52)$$

with removable interpretation at $t = 0$ and locally uniform convergence in $t \in \mathbb{C}$. Integration by parts also gives the bridge

$$\Phi(t) = e^{-it} + 2ite^{it} A(2t). \quad (53)$$

Replacing t by $-is$ turns (52) into an entire Laplace series with sinhc^2 kernels.

3.6 Exact-rational inverse interpolants

Let $G = F^{-1} : [0, 1] \rightarrow [0, 1]$. The inverse-frontier manuscripts in the pinned corpus develop sharp endpoint reversion, inverse derivatives, and quantile transforms. The Faber construction supplies a complementary finite exact representation.

For $J \geq 0$ and $0 \leq k \leq 2^J$, put

$$y_{J,k} = F(k/2^J), \quad d_{J,k} = y_{J,k+1} - y_{J,k} > 0 \quad (0 \leq k < 2^J). \quad (54)$$

Every $y_{J,k}$ and $d_{J,k}$ is rational. Define G_J on $[y_{J,k}, y_{J,k+1}]$ by

$$G_J(y) = \frac{k}{2^J} + \frac{2^{-J}}{d_{J,k}}(y - y_{J,k}). \quad (55)$$

Equivalently, with $s_{J,k} = 2^{-J}/d_{J,k}$,

$$G_J(y) = s_{J,0}y + \sum_{k=1}^{2^J-1} (s_{J,k} - s_{J,k-1})(y - y_{J,k})_+. \quad (56)$$

Theorem 3.8 (Exact-rational inverse limit). *The map G_J is the inverse of the Faber partial sum $S_J F$. It is a strictly increasing piecewise-affine homeomorphism with rational knots, rational slopes, and the reflection law*

$$G_J(1 - y) = 1 - G_J(y). \quad (57)$$

Moreover,

$$G(y) = \lim_{J \rightarrow \infty} G_J(y) \quad (58)$$

uniformly on $[0, 1]$. If

$$\omega_G(\delta) = \sup_{|u-v| \leq \delta} |G(u) - G(v)|, \quad (59)$$

then the explicit estimate

$$\|G_J - G\|_{\infty} \leq \omega_G(4^{-J}) \quad (60)$$

holds.

Proof. The dyadic node values of $S_J F$ are $y_{J,k}$, and strict monotonicity of F makes every chord slope positive. Thus (55) is exactly $(S_J F)^{-1}$; (56) is the standard slope-jump form of the same piecewise-affine map. Reflection of the dyadic node table proves (57).

Fix y and put $x = G_J(y)$. Since $S_J F(x) = y$ and $\|S_J F - F\|_\infty \leq 4^{-J}$,

$$y - 4^{-J} \leq F(x) \leq y + 4^{-J},$$

with the endpoints clipped to $[0, 1]$. Applying the increasing map G places x between the corresponding two quantiles, which gives (60). Uniform continuity of G completes the proof. $\square$

New deduction relative to the pinned corpus. Equations (55) and (56) give a finite exact-rational neural-spline or truncated-power representation of the inverse at every level. The sharp Lambert-periodic endpoint expansion already in the corpus may be inserted into $\omega_G(4^{-J})$ to resolve the non-algebraic global convergence rate.

4 Rational Haar expansions and a Haar–sinc transform

4.1 The derivative as a dyadic martingale limit

Define the unnormalized Haar step

$$H_{m,k}(x) = \mathbf{1}_{[\ell_{m,k}, \xi_{m,k})}(x) - \mathbf{1}_{[\xi_{m,k}, r_{m,k})}(x). \quad (61)$$

Then $\Lambda'_{m,k} = 2^{m+1} H_{m,k}$ away from its three corners.

Theorem 4.1 (Exact rational Haar representation). *The derivative of the Fabius function has the Haar expansion*

$$F'(x) = 1 + \sum_{m=0}^{\infty} \sum_{k=0}^{2^m-1} 2^{m+1} c_{m,k} H_{m,k}(x). \quad (62)$$

The partial sum through $m = J - 1$ is the derivative of $S_J F$ and equals the cell average of F' on every dyadic cell of level J . It converges to F' in $L^p(0, 1)$ for every $1 \leq p \leq \infty$, where $p = \infty$ is understood in the essential-supremum sense.

Proof. Differentiate the finite identity (31). On a cell $[a, b]$ of the level- J mesh, the derivative of the interpolant is $[F(b) - F(a)]/(b - a)$, which is the average of F' on that cell. These conditional averages converge uniformly to the continuous function F' by uniform continuity. Values at the finitely many grid corners are irrelevant in every L^p space. $\square$

Combining (62) with (4) yields

$$\text{up}(2x - 1) = \frac{1}{2} + \sum_{m=0}^{\infty} \sum_{k=0}^{2^m-1} 2^m c_{m,k} H_{m,k}(x). \quad (63)$$

Thus the compact C^∞ bump is an L^∞ limit of exact rational step functions. The apparent tension is useful: smoothness is encoded by strong cancellation across infinitely many dyadic scales.

4.2 Fourier transform of one Haar cell

A direct calculation gives

$$\int_0^1 H_{m,k}(x) e^{-itx} dx = e^{-itk/2^m} \frac{(1 - e^{-it/2^{m+1}})^2}{it}. \quad (64)$$

Apply this to (63), remembering that the centered coordinate is $y = 2x - 1$.

Theorem 4.2 (Haar–sinc² expansion of the sinc product). *For every $t \in \mathbb{C}$,*

$$\Phi(t) = \text{sinc } t + \sum_{m=0}^{\infty} \sum_{k=0}^{2^m-1} \frac{2^m c_{m,k}}{it} e^{it(1-2k/2^m)} \left(1 - e^{-it/2^m}\right)^2, \quad (65)$$

$$= \text{sinc } t - \sum_{m=0}^{\infty} \sum_{k=0}^{2^m-1} \frac{2^{m+2} c_{m,k}}{it} e^{-ity_{m,k}} \sin^2\left(\frac{t}{2^{m+1}}\right), \quad (66)$$

where

$$y_{m,k} = \frac{2k+1}{2^m} - 1. \quad (67)$$

Every apparent singularity at $t = 0$ is removable, and both series converge normally on compact subsets of $\mathbb{C}$.

Proof. Use

$$\Phi(t) = 2e^{it} \int_0^1 \text{up}(2x-1) e^{-2itx} dx$$

and insert the finite Haar partial sums. Equation (64) with frequency $2t$ gives (65); factoring $(1 - e^{-ia})^2 = -4e^{-ia} \sin^2(a/2)$ gives (66). For t in a fixed compact set, (35) and $|\sin(t/2^{m+1})| \leq C_K 2^{-m}$ bound the entire level- m contribution by $C'_K 4^{-m}$. The Weierstrass test proves normal convergence. $\square$

Pairing reflected coefficients gives a manifestly real formula on the real axis.

Corollary 4.3 (Real paired form). *For real t ,*

$$\Phi(t) = \text{sinc } t - \sum_{m=1}^{\infty} \frac{2^{m+3}}{t} \sin^2\left(\frac{t}{2^{m+1}}\right) \sum_{k=0}^{2^{m-1}-1} c_{m,k} \sin\left(t \left[1 - \frac{2k+1}{2^m}\right]\right). \quad (68)$$

The representation (68) is a series–product bridge. The left side is the infinite sinc product; the right side consists of exact rational curvature coefficients multiplied by elementary sine and $\sin^2$ factors. It gives a new way to transfer arithmetic information from dyadic values of F into the Fourier image.

5 A positive rational series for the sinc energy

5.1 Haar Parseval

Let

$$A_2 = \int_0^1 F(x)^2 dx. \quad (69)$$

The audited integral report identifies the same constant with a sinc-product energy, but leaves its arithmetic nature open. The Haar representation gives a monotone exact-rational decomposition.

Theorem 5.1 (Positive rational energy series). *One has*

$$A_2 = \frac{1}{4} + \sum_{m=0}^{\infty} 2^m \sum_{k=0}^{2^m-1} c_{m,k}^2. \quad (70)$$

Every summand is a nonnegative rational number. Hence the partial sums

$$R_J = \frac{1}{4} + \sum_{m=0}^{J-1} 2^m \sum_{k=0}^{2^m-1} c_{m,k}^2 \quad (71)$$

are rational and increase monotonically to A_2 .

Proof. The Haar steps are pairwise orthogonal, $\|H_{m,k}\|_2^2 = 2^{-m}$, and all have zero mean. Parseval applied to (62) yields

$$\int_0^1 F'(x)^2 dx = 1 + \sum_{m=0}^{\infty} 2^{m+2} \sum_{k=0}^{2^m-1} c_{m,k}^2. \quad (72)$$

On the other hand, $F'(x) = 2 \operatorname{up}(2x - 1)$ and evenness give

$$\int_0^1 F'^2 = 2 \int_{-1}^1 \operatorname{up}(y)^2 dy = 4 \int_0^1 F(x)^2 dx = 4A_2.$$

Divide (72) by four. □

Angular Parseval and (14) give the three-way identity

$$\frac{1}{4} + \sum_{m \geq 0} 2^m \sum_k c_{m,k}^2 = \int_0^1 F(x)^2 dx = \frac{1}{2\pi} \int_0^\infty \prod_{j=1}^\infty \operatorname{sinc}^2\left(\frac{t}{2^j}\right) dt. \quad (73)$$

Thus an improper integral of a transcendental infinite product is represented by a positive series of exact rational squares.

Corollary 5.2 (Certified elementary tail bound). *For every $J \geq 0$,*

$$0 < A_2 - R_J \leq \frac{4}{3} 4^{-J}. \quad (74)$$

Proof. By (35), the level- m energy is at most

$$2^m \cdot 2^m \cdot 2^{-4m} = 4^{-m}.$$

Sum the geometric tail from $m = J$. □

5.2 Derivative norm scaling

Put

$$I_r = \int_0^1 \left(F^{(r)}(x)\right)^2 dx, \quad r \geq 1. \quad (75)$$

The derivative tiling already present in the corpus implies the exact norm recurrence $I_r = 4^r I_{r-1}$. Since $I_1 = 4A_2$, this gives

$$I_r = 4^{r(r+1)/2} A_2. \quad (76)$$

We use this baseline identity to evaluate every coefficient in the new energy-tail expansion.

5.3 All-orders level asymptotics

Define the contribution of one dyadic scale by

$$E_m = 2^m \sum_{k=0}^{2^m-1} c_{m,k}^2, \quad A_2 = \frac{1}{4} + \sum_{m \geq 0} E_m. \quad (77)$$

Theorem 5.3 (Complete asymptotic expansion of one level). *As $m \rightarrow \infty$,*

$$E_m \sim A_2 \sum_{q=1}^{\infty} \beta_q 4^{-qm}, \quad (78)$$

where

$$\beta_q = (-1)^{q+1} \frac{2^{q(q+1)+1} (2^{2q} - 1)}{(2q+2)!}. \quad (79)$$

In particular,

$$\beta_1 = 1, \quad \beta_2 = -\frac{8}{3}, \quad \beta_3 = \frac{64}{5}, \quad \beta_4 = -\frac{139264}{945}. \quad (80)$$

Proof. A detailed remainder proof is given in [section A](#); here is the coefficient calculation. Write $\Delta = 2^{-m}$ and $h = \Delta/2$. At a cell center,

$$c_{m,k} = -(\cosh(hD) - 1)F(\xi_{m,k}), \quad D = \frac{d}{dx},$$

in the sense of an arbitrary finite Taylor expansion with uniform remainder. The midpoint Riemann sum differs from the corresponding integral by a superalgebraically small quantity because all derivatives entering the integrand are flat at both endpoints. Hence

$$E_m \sim \Delta^{-2} \int_0^1 [(\cosh(hD) - 1)F(x)]^2 dx.$$

Now

$$(\cosh z - 1)^2 = \sum_{r=2}^{\infty} \frac{2^{2r-1} - 2}{(2r)!} z^{2r}. \quad (81)$$

Repeated integration by parts gives $\int F D^{2r} F = (-1)^r I_r$. Substituting $h = \Delta/2$, setting $q = r - 1$, and using (76) produces (79). $\square$

Corollary 5.4 (Leading level constant).

$$\lim_{m \rightarrow \infty} 4^m E_m = A_2. \quad (82)$$

The same constant therefore controls the original integral, the total Haar energy, and the leading asymptotic energy at every sufficiently fine scale. This self-referential feature is a consequence of the exact derivative norm scaling.

5.4 All-orders tail and acceleration

Summing (78) geometrically gives the tail expansion.

Corollary 5.5 (Complete rational tail expansion). *For $J \rightarrow \infty$,*

$$A_2 - R_J \sim A_2 \sum_{q=1}^{\infty} \gamma_q 4^{-qJ}, \quad (83)$$

where

$$\gamma_q = \frac{\beta_q}{1 - 4^{-q}}. \quad (84)$$

The first coefficients are

$$\gamma_1 = \frac{4}{3}, \quad \gamma_2 = -\frac{128}{45}, \quad \gamma_3 = \frac{4096}{315}, \quad \gamma_4 = -\frac{2097152}{14175}. \quad (85)$$

In particular,

$$\boxed{\lim_{J \rightarrow \infty} 4^J (A_2 - R_J) = \frac{4}{3} A_2.} \quad (86)$$

For any fixed truncation order Q , define

$$A_2^{[Q]}(J) = \frac{R_J}{1 - \sum_{q=1}^Q \gamma_q 4^{-qJ}}. \quad (87)$$

Then

$$A_2^{[Q]}(J) - A_2 = \mathcal{O}(4^{-(Q+1)J}) \quad (88)$$

for fixed Q . The numerator and denominator in (87) are rational, so every accelerated value is itself an exact rational number before decimal conversion.

Exact/numerical check. Exact level-16 data and seven tail coefficients give

$$A_2 \approx 0.404436767983985281465621491145376633448000519787051529947065. \quad (89)$$

The stable digits agree across neighboring values of J and Q , and the raw level-16 rational partial sum is 0.4044367678584316 The numerical table and convergence plot in [section 12](#) show the acceleration explicitly.

6 Triangular Bernoulli and independent-binomial layers

6.1 Regrouping the binary digits

The uniform-coordinate representation (6) hides a second independence structure. Choose the nonterminating binary expansion only on the null set where two choices exist, and write

$$U_j = \sum_{r=1}^{\infty} B_{j,r} 2^{-r}, \quad (90)$$

where all $B_{j,r}$ are independent Bernoulli(1/2) variables. The weight of $B_{j,r}$ in X depends only on $n = j + r$.

Theorem 6.1 (Independent triangular binomial decomposition). *There exist independent random variables*

$$K_n \sim \text{Binomial}(n - 1, \tfrac{1}{2}), \quad n \geq 2, \quad (91)$$

such that

$$\boxed{X = \sum_{n=2}^{\infty} \frac{K_n}{2^n}.} \quad (92)$$

Equivalently, if S_{n-1} is a sum of $n - 1$ independent Rademacher signs and the sums are independent across n , then

$$Y = 2X - 1 = \sum_{n=2}^{\infty} \frac{S_{n-1}}{2^n}. \quad (93)$$

Proof. Substitute (90) into $X = \sum_{j \geq 1} 2^{-j} U_j$ and rearrange the absolutely convergent nonnegative series:

$$X = \sum_{j,r \geq 1} B_{j,r} 2^{-(j+r)} = \sum_{n \geq 2} 2^{-n} \sum_{j=1}^{n-1} B_{j,n-j}.$$

For fixed n the inner sum has the law in (91); different n use disjoint sets of bits and are independent. Centering each binomial sum proves (93). $\square$

This is not merely another proof of the random-series representation. It changes the elementary building blocks from one uniform variable per scale to one increasingly concentrated binomial variable per antidiagonal of the binary digit array.

6.2 A triangular cosine product

Corollary 6.2 (Product–product identity). *The moment-generating function and centered Fourier transform factor as*

$$M_X(s) = \prod_{n=2}^{\infty} \left(\frac{1 + e^{s/2^n}}{2} \right)^{n-1}. \quad (94)$$

$$\Phi(t) = \prod_{n=2}^{\infty} \cos\left(\frac{t}{2^n}\right)^{n-1} = \prod_{j=1}^{\infty} \operatorname{sinc}\left(\frac{t}{2^j}\right). \quad (95)$$

Both products converge normally on compact subsets of $\mathbb{C}$.

Proof. The first two factorizations follow from independence in theorem 6.1. For the last equality, use the classical binary product $\operatorname{sinc} z = \prod_{r \geq 1} \cos(z/2^r)$ and regroup the normally convergent double product:

$$\prod_{j \geq 1} \operatorname{sinc}(t/2^j) = \prod_{j,r \geq 1} \cos(t/2^{j+r}) = \prod_{n \geq 2} \cos(t/2^n)^{n-1}.$$

$\square$

The logarithmic identity

$$\log \Phi(t) = \sum_{n=2}^{\infty} (n-1) \log \cos(t/2^n) \quad (96)$$

reorganizes every cumulant as a weighted geometric sum. It also offers a different truncation strategy: the exponent $n - 1$ concentrates many tiny Rademacher increments into one layer.

Let

$$\Phi_N^{\triangle}(t) = \prod_{n=2}^N \cos(t/2^n)^{n-1}. \quad (97)$$

The omitted factor is quantitatively controlled.

Proposition 6.3 (Local triangular-product tail). *For $|t| \leq 2^N$, write $\Phi(t) = \Phi_N^{\triangle}(t) T_N(t)$, where $T_N(t) > 0$ on the real axis. Then*

$$0 \leq -\log T_N(t) \leq \frac{(3N+1)t^2}{9 \cdot 4^N}. \quad (98)$$

Proof. For $n > N$, $|t|/2^n \leq 1/2$, and $0 \leq -\log \cos u \leq u^2$ on $|u| \leq 1/2$. Therefore

$$-\log T_N(t) \leq t^2 \sum_{n>N} (n-1)4^{-n} = \frac{(3N+1)t^2}{9 \cdot 4^N}.$$

□

6.3 Finite combinatorial CDFs

Define

$$X_N^\triangle = \sum_{n=2}^N \frac{K_n}{2^n}, \quad \delta_N = \sum_{n>N} \frac{n-1}{2^n} = \frac{N+1}{2^N}. \quad (99)$$

The tail lies in $[0, \delta_N]$, so

$$\mathbb{P}(X_N^\triangle \leq x - \delta_N) \leq F(x) \leq \mathbb{P}(X_N^\triangle \leq x). \quad (100)$$

Expanding the independent binomial laws gives the finite sum

$$F_N^\triangle(x) = 2^{-N(N-1)/2} \sum_{k_2=0}^1 \cdots \sum_{k_N=0}^{N-1} \left(\prod_{n=2}^N \binom{n-1}{k_n} \right) \mathbf{1}_{\{\sum_{n=2}^N k_n 2^{-n} \leq x\}}. \quad (101)$$

Then $F_N^\triangle(x) \rightarrow F(x)$ at every real x .

The complete mass distribution is encoded by the integer polynomial

$$P_N(q) = \prod_{n=2}^N \left(1 + q^{2^{N-n}} \right)^{n-1}. \quad (102)$$

Its degree and total coefficient sum are

$$\deg P_N = 2^N - N - 1, \quad P_N(1) = 2^{N(N-1)/2}. \quad (103)$$

If $p_{N,\ell} = [q^\ell]P_N(q)$, then

$$\mathbb{P}(2^N X_N^\triangle = \ell) = 2^{-N(N-1)/2} p_{N,\ell}, \quad (104)$$

and hence

$$F(x) = \lim_{N \rightarrow \infty} 2^{-N(N-1)/2} \sum_{0 \leq \ell \leq 2^N x} p_{N,\ell}. \quad (105)$$

This is a limit of cumulative coefficient sums of explicit sparse products. It is complementary to the alternating Thue–Morse spline (20): one finite model is signed and polynomial in x ; the other is positive and combinatorial on a dyadic lattice.

Every factor in (102) is palindromic, so the coefficient sequence is symmetric. Computation suggests substantially more.

Conjecture 6.4 (Triangular log-concavity). *For every $N \geq 2$, the coefficients of $P_N(q)$ are log-concave:*

$$p_{N,\ell}^2 \geq p_{N,\ell-1} p_{N,\ell+1} \quad (1 \leq \ell < 2^N - N - 1). \quad (106)$$

Consequently they are unimodal.

Exact/numerical check. The bundled program verifies (106) in exact integer arithmetic for every $2 \leq N \leq 16$. The largest checked polynomial has degree 65519 and total mass 2^{120} . This is evidence, not a proof: sparse dilation products do not inherit log-concavity from their factors by a routine closure theorem.

7 Odd-lattice reconstruction of the Fourier image

7.1 Paley–Wiener sampling

Because $up \in C_c^\infty([-1, 1])$, its transform satisfies

$$\Phi \in \text{PW}_1 \cap \mathcal{S}(\mathbb{R}) : \quad (107)$$

Φ is entire of exponential type at most one, square-integrable on the real axis, and rapidly decreasing there. The Whittaker–Shannon cardinal sampling theorem on PW_1 [10, 11, 12] gives

$$\Phi(z) = \sum_{k \in \mathbb{Z}} \Phi(k\pi) \frac{\sin(z - k\pi)}{z - k\pi}. \quad (108)$$

Rapid decay of the samples strengthens the usual L^2 statement to absolute and locally uniform convergence in $z \in \mathbb{C}$.

All nonzero even samples vanish because $\Phi(2\pi n) = 0$. Put

$$s_m = \Phi((2m+1)\pi) = \prod_{j=1}^{\infty} \text{sinc}\left(\frac{(2m+1)\pi}{2^j}\right). \quad (109)$$

Theorem 7.1 (Odd-sample cardinal representation). *For all $z \in \mathbb{C}$,*

$$\Phi(z) = \frac{\sin z}{z} - \sin z \sum_{m \in \mathbb{Z}} \frac{\Phi((2m+1)\pi)}{z - (2m+1)\pi}, \quad (110)$$

$$\Phi(z) = \frac{\sin z}{z} - 2z \sin z \sum_{m=0}^{\infty} \frac{s_m}{z^2 - (2m+1)^2 \pi^2}. \quad (111)$$

At the apparent poles the expressions are interpreted by their removable limits. Both series converge normally on compact subsets of $\mathbb{C}$ after those removals.

Proof. In (108), the term $k = 0$ is $\sin z/z$ and every nonzero even term vanishes. For odd k , $\sin(z - k\pi) = -\sin z$, giving (110). Pair the samples at $\pm(2m+1)\pi$ and use evenness of Φ to obtain (111). $\square$

New deduction relative to the pinned corpus. Equation (19) reconstructs the physical bump from the same odd samples. Equation (111) instead reconstructs the entire Fourier image at arbitrary complex arguments. Together they make the odd sample sequence a two-sided coordinate system for both domains.

For every compact $K \subset \mathbb{C}$, rapid sample decay implies the superalgebraic truncation estimate

$$\sup_{z \in K} |\Phi(z) - \Phi_M(z)| = \mathcal{O}_{K,A}(M^{-A}) \quad \text{for every } A > 0, \quad (112)$$

where Φ_M retains $0 \leq m < M$ in the paired series.

7.2 Higher zero multiplicities as odd-sample sum rules

Factor (111) as

$$\Phi(z) = \sin z B(z), \quad (113)$$

where, away from its removable odd-lattice singularities,

$$B(z) = \frac{1}{z} - 2z \sum_{m=0}^{\infty} \frac{s_m}{z^2 - (2m+1)^2 \pi^2}. \quad (114)$$

At an even lattice point $z_n = 2\pi n \neq 0$, B is analytic. The established zero order $1 + \nu_2(n)$ of Φ now becomes a hierarchy of identities among the odd samples.

Theorem 7.2 (Odd-sample zero-jet hierarchy). *For every nonzero integer n ,*

$$B^{(r)}(2\pi n) = 0 \quad (0 \leq r < \nu_2(n)), \quad B^{(\nu_2(n))}(2\pi n) \neq 0. \quad (115)$$

In particular, for every positive even n ,

$$\sum_{m=0}^{\infty} \frac{s_m}{4n^2 - (2m+1)^2} = \frac{1}{8n^2}. \quad (116)$$

If $4 \mid n$, then also

$$\sum_{m=0}^{\infty} \frac{s_m}{(4n^2 - (2m+1)^2)^2} = \frac{1}{32n^4}. \quad (117)$$

Proof. The sine factor has a simple zero at $2\pi n$, while Φ has order $1 + \nu_2(n)$; hence B has order $\nu_2(n)$ there. This proves (115). Setting $B(2\pi n) = 0$ and dividing out powers of π gives (116). Differentiate (114):

$$B'(z) = -\frac{1}{z^2} - 2 \sum_m \frac{s_m}{z^2 - a_m^2} + 4z^2 \sum_m \frac{s_m}{(z^2 - a_m^2)^2}, \quad a_m = (2m+1)\pi.$$

If $4 \mid n$, then $B'(2\pi n) = 0$. Substituting (116) yields (117). $\square$

Higher derivatives of B give explicit rational combinations of

$$\sum_{m \geq 0} \frac{s_m}{(4n^2 - (2m+1)^2)^j}, \quad j = 1, 2, \dots,$$

whenever $2^r \mid n$. This translates the 2-adic zero multiplicity into a nested family of exact cardinal-sample identities. It is a promising bridge between the spectral arithmetic already formalized in Lean and classical sampling theory.

8 Chebyshev, Gegenbauer, and Bessel–product expansions

8.1 Chebyshev coefficients

Write the Chebyshev expansion in the standard form

$$\text{up}(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n T_n(x), \quad -1 \leq x \leq 1, \quad (118)$$

where

$$a_n = \frac{2}{\pi} \int_{-1}^1 \frac{\text{up}(x) T_n(x)}{\sqrt{1-x^2}} dx. \quad (119)$$

Since $\text{up}(\cos \theta)$ is a smooth 2π -periodic even function, the series converges absolutely and uniformly, and its coefficients decay faster than every power.

The plane-wave identity

$$\int_{-1}^1 \frac{e^{itx} T_n(x)}{\sqrt{1-x^2}} dx = \pi i^n J_n(t) \quad (120)$$

turns the coefficients into transforms of the sinc product.

Theorem 8.1 (Chebyshev–Bessel sinc representation). *All odd coefficients vanish, and for $r \geq 0$,*

$$a_{2r} = \frac{2(-1)^r}{\pi} \int_0^\infty J_{2r}(t) \prod_{j=1}^\infty \operatorname{sinc}\left(\frac{t}{2^j}\right) dt. \quad (121)$$

The integral is absolutely convergent.

Proof. Insert Fourier inversion (17) into (119), apply Fubini, and use (120). This gives

$$a_n = \frac{i^n}{\pi} \int_{-\infty}^\infty \Phi(t) J_n(t) dt.$$

Parity kills odd n and doubles the positive half-axis for even n . Rapid decay of Φ makes the integral absolute. $\square$

The first numerical coefficients are approximately

$$\begin{aligned} a_0 &= 0.681978961331094\dots, \\ a_2 &= -0.514948055352838\dots, \\ a_4 &= 0.195935940418300\dots, \\ a_6 &= -0.002356862179497\dots, \\ a_8 &= -0.028210793776951\dots, \\ a_{10} &= 0.011427972423053\dots \end{aligned} \quad (122)$$

The sign change between a_6 and a_8 shows that a naive strict alternation conjecture would already be false.

8.2 A Gegenbauer family

Let $\lambda > 0$, let C_n^λ be the Gegenbauer polynomial, and put

$$h_n^{(\lambda)} = \int_{-1}^1 (1-x^2)^{\lambda-1/2} [C_n^\lambda(x)]^2 dx = \frac{\pi 2^{1-2\lambda} \Gamma(n+2\lambda)}{n!(n+\lambda)\Gamma(\lambda)^2}. \quad (123)$$

Write

$$\operatorname{up}(x) = \sum_{n=0}^\infty a_n^{(\lambda)} C_n^\lambda(x), \quad a_n^{(\lambda)} = \frac{1}{h_n^{(\lambda)}} \int_{-1}^1 \operatorname{up}(x) C_n^\lambda(x) (1-x^2)^{\lambda-1/2} dx. \quad (124)$$

Define

$$K_{n,\lambda} = 2^\lambda \sqrt{\pi} \Gamma(\lambda + \tfrac{1}{2}) \frac{(2\lambda)_n}{n!}. \quad (125)$$

Theorem 8.2 (Gegenbauer–Bessel family). *The odd coefficients vanish, and*

$$a_{2r}^{(\lambda)} = \frac{(-1)^r K_{2r,\lambda}}{\pi h_{2r}^{(\lambda)}} \int_0^\infty \Phi(t) t^{-\lambda} J_{2r+\lambda}(t) dt. \quad (126)$$

Proof. Use the plane-wave integral

$$\int_{-1}^1 e^{itx} C_n^\lambda(x) (1-x^2)^{\lambda-1/2} dx = i^n K_{n,\lambda} t^{-\lambda} J_{n+\lambda}(t)$$

after interpreting the right side by its entire normalized-Bessel continuation at $t = 0$. Fourier inversion and parity give (126). $\square$

The choice $\lambda = 1/2$ recovers Legendre polynomials; standard normalization facts are as in [13]. If

$$L_{2r} = \frac{4r+1}{2} \int_{-1}^1 \text{up}(x) P_{2r}(x) dx \in \mathbb{Q} \quad (127)$$

is the exact rational coefficient from the established Fourier–Legendre expansion, and j_n is the spherical Bessel function, then

$$L_{2r} = \frac{(4r+1)(-1)^r}{\pi} \int_0^\infty \Phi(t) j_{2r}(t) dt. \quad (128)$$

Thus the corpus’s rational Legendre coefficients evaluate an infinite family of Bessel–sinc-product integrals exactly:

$$\int_0^\infty j_{2r}(t) \prod_{j \geq 1} \text{sinc}(t/2^j) dt = \frac{(-1)^r \pi}{4r+1} L_{2r}. \quad (129)$$

9 Exact-rational Bernstein limits and Kummer transforms

9.1 Dyadic Bernstein polynomials for F

Bernstein’s probabilistic approximation scheme [9] starts, for $n \geq 1$, with

$$(B_n F)(x) = \sum_{k=0}^n F(k/n) \binom{n}{k} x^k (1-x)^{n-k}. \quad (130)$$

When $n = 2^r$, every sample $F(k/n)$ is rational. Hence $B_{2^r} F$ is an exact-rational polynomial in either the Bernstein or monomial basis.

Theorem 9.1 (Rational Bernstein limit with explicit error). *For $n = 2^r$,*

$$F(x) = \lim_{r \rightarrow \infty} (B_{2^r} F)(x) \quad (131)$$

uniformly on $[0, 1]$. More precisely,

$$|B_n F(x) - F(x)| \leq \frac{\|F''\|_\infty}{2n} x(1-x) \leq \frac{1}{n}. \quad (132)$$

Proof. If $K \sim \text{Binomial}(n, x)$, then $B_n F(x) = \mathbb{E}F(K/n)$. Taylor’s theorem about x , the zero mean of $K/n - x$, and $\mathbb{E}(K/n - x)^2 = x(1-x)/n$ give the first bound. Use $\|F''\|_\infty = 8$ and $x(1-x) \leq 1/4$ for the second. $\square$

This is a different exact-rational limit from the Faber expansion. Faber partial sums interpolate all dyadic nodes and converge like 4^{-J} when the mesh is 2^{-J} ; Bernstein polynomials smooth the node data by a binomial kernel and converge at the slower but shape-preserving order 2^{-J} .

9.2 Finite Kummer sums for transforms

The beta–Kummer integral is

$$\int_0^1 \binom{n}{k} x^k (1-x)^{n-k} e^{zx} dx = \frac{1}{n+1} {}_1F_1(k+1; n+2; z). \quad (133)$$

Consequently the restricted Fabius transform $A(t)$ from (50) has the locally uniform entire limit

$$A(t) = \lim_{r \rightarrow \infty} \frac{1}{2^r + 1} \sum_{k=0}^{2^r} F(k/2^r)_1 F_1(k+1; 2^r+2; -it). \quad (134)$$

Every coefficient in every finite sum is rational.

For the bump, put

$$g(u) = \text{up}(2u-1) = F(1-|2u-1|), \quad 0 \leq u \leq 1. \quad (135)$$

At dyadic nodes, $g(k/2^r) \in \mathbb{Q}$. Approximate g by $B_n g$ and transport back to $[-1, 1]$. Its Fourier image is the finite sum

$$\Phi_n(t) = \frac{2e^{it}}{n+1} \sum_{k=0}^n g(k/n)_1 F_1(k+1; n+2; -2it). \quad (136)$$

Theorem 9.2 (Bernstein–Kummer representation of the sinc product). *Along dyadic degrees $n = 2^r$,*

$$\prod_{j=1}^{\infty} \text{sinc}\left(\frac{t}{2^j}\right) = \lim_{r \rightarrow \infty} \frac{2e^{it}}{2^r + 1} \sum_{k=0}^{2^r} F(1-|2k/2^r-1|)_1 F_1(k+1; 2^r+2; -2it). \quad (137)$$

The convergence is locally uniform in $t \in \mathbb{C}$.

Proof. Bernstein’s theorem gives $B_n g \rightarrow g$ uniformly. On a compact set $K \subset \mathbb{C}$, the Fourier kernels on $[-1, 1]$ are bounded by $\exp(\sup_{t \in K} |\text{Im } t|)$, so uniform approximation of the physical functions implies locally uniform convergence of their transforms. Equation (133) evaluates each finite transform exactly. $\square$

This representation combines a limit, a finite hypergeometric series, and exact dyadic arithmetic. It is well suited to symbolic experimentation: for each fixed r the entire approximant satisfies the differential relations of a finite family of Kummer functions, while the limit is known from the corpus to be non-holonomic in much stronger senses.

10 Weighted cube volumes and hyperplane sections

10.1 Sublevel volumes

Let $a_j = 2^{-j}$ and $a^{(N)} = (a_1, \dots, a_N)$. The finite CDF in (20) is exactly a weighted-cube sublevel volume:

$$F_N(x) = \text{vol}_N \left\{ u \in [0, 1]^N : \sum_{j=1}^N a_j u_j \leq x \right\}. \quad (138)$$

Therefore

$$F(x) = \lim_{N \rightarrow \infty} \text{vol}_N \left\{ u \in [0, 1]^N : \sum_{j=1}^N 2^{-j} u_j \leq x \right\} \quad (139)$$

for every real x . Equation (20) is the inclusion–exclusion evaluation of the same polytope volume.

One may also write the infinite-dimensional product-measure integral

$$F(x) = \int_{[0,1]^{\mathbb{N}}} \mathbf{1}_{\{\sum_{j \geq 1} 2^{-j} u_j \leq x\}} d\mathbf{u}. \quad (140)$$

This is the geometric form of the probability model, but it emphasizes that the Fabius function is the distribution of weighted half-spaces cutting a Hilbert cube.

10.2 Hyperplane sections and a Fourier-slice limit

Let v be uniform on $[-1, 1]^N$ and set

$$Y_N = a^{(N)} \cdot v. \quad (141)$$

By the coarea formula, its density u_N is

$$u_N(y) = \frac{2^{-N}}{\|a^{(N)}\|_2} \mathcal{H}^{N-1} \left\{ v \in [-1, 1]^N : a^{(N)} \cdot v = y \right\}. \quad (142)$$

The Fourier-slice theorem reduces here to Fubini:

$$\int_{\mathbb{R}} u_N(y) e^{-ity} dy = \prod_{j=1}^N \text{sinc}(t/2^j). \quad (143)$$

The first three sinc factors give an integrable majorant for all later finite products, so dominated Fourier inversion yields uniform convergence $u_N \rightarrow \text{up}$. Consequently

$$\text{up}(y) = \lim_{N \rightarrow \infty} \frac{2^{-N}}{\|a^{(N)}\|_2} \mathcal{H}^{N-1} \left\{ v \in [-1, 1]^N : a^{(N)} \cdot v = y \right\}. \quad (144)$$

Equations (139), (144), and (143) form a volume–section–product triangle: sublevel volumes give F , codimension-one sections give up, and Fourier slices give the finite sinc products converging to Φ .

11 Representation synthesis

The formulas above are best viewed as a network rather than a list. Exact dyadic arithmetic supplies Faber coefficients; differentiation turns them into Haar coefficients; Haar Parseval gives a positive energy series; Haar Fourier transforms give a new sinc-product series. Independently, binary digits reorganize the same probability law into binomial layers, while compact support places the Fourier image in a sampling space. Orthogonal polynomials and Bernstein polynomials then provide two different spectral and algebraic envelopes.

Table 3: New representation families and their characteristic data.

Representation	Formula type	Exact data	Convergence/control
Faber–Schauder	$F = x + \sum c_{m,k} \Lambda_{m,k}$	Rational dyadic midpoint defects	Uniform; sharp error $\sim 4^{-J}$.
Inverse splines	$G_J = (S_J F)^{-1}$, formula (56)	Rational knots and slopes	Uniform; error $\leq \omega_G(4^{-J})$.
Rvachev tents	$\text{up}(y) = 1 - y + \sum c \Lambda(1 - y)$	Same rational table	Uniform on support.
Haar density	$F' = 1 + \sum 2^{m+1} c H$	Rational step coefficients	L^p , including essential L^∞ .
Haar Fourier	$\Phi = \text{sinc} + \sum c e^{i \cdot} \sin^2 / (it)$	Rational coefficients, elementary kernels	Normal on compact subsets of $\mathbb{C}$.
Energy	$A_2 = 1/4 + \sum 2^m \sum c^2$	Positive rational squares	Monotone, certified geometric tail.
Energy asymptotic	$E_m \sim A_2 \sum \beta_q 4^{-qm}$	Explicit rational β_q, γ_q	Poincaré all orders.
Triangular	$X = \sum 2^{-n} K_n$	Independent binomial layers	Almost sure and L^p .
probability			
Triangular product	$\Phi = \prod \cos(t/2^n)^{n-1}$	Integer exponents	Normal on compacta; local tail bound.

Representation	Formula type	Exact data	Convergence/control
Coefficient CDF	Cumulative coefficients of $P_N(q)$	Positive integers	Bracket width $(N+1)2^{-N}$.
Odd cardinal	Rational-kernel sum over $s_m = \Phi((2m+1)\pi)$	Explicit sinc products	Locally uniform, superalgebraic truncation.
Sample sum rules	Zero jets of $B(z)$	2-adic multiplicities	Exact convergent identities.
Chebyshev–Gegenbauer	Bessel integrals of Φ	Existing moments/Legendre rationals	Absolute integrals; uniform polynomial series.
Bernstein	Exact-rational polynomials at dyadic degrees	Dyadic values of F	Uniform; explicit $1/n$ bound for F .
Kummer Fourier	Finite ${}_1F_1$ sums, then $n \rightarrow \infty$	Rational finite weights	Locally uniform entire convergence.
Cube geometry	Sublevel volumes and hyperplane sections	Dyadic normal vector	CDF and uniform-density limits.

Several hybrid formulas deserve emphasis:

$$\text{rational wavelets} + \text{elementary oscillation} \longrightarrow \text{entire sinc product}, \quad (145)$$

$$\text{odd sinc-product samples} + \text{rational resolvents} \longrightarrow \text{entire sinc product}, \quad (146)$$

$$\text{rational dyadic samples} + {}_1F_1 \longrightarrow \text{entire sinc product}, \quad (147)$$

$$\text{rational Legendre coefficients} \longleftrightarrow \text{Bessel–sinc integrals}. \quad (148)$$

The same function is therefore reconstructed from local curvature, stepwise martingale data, odd spectral samples, binomial coefficient arrays, orthogonal-polynomial periods, and weighted-polytope geometry.

12 Numerical and exact verification

12.1 Reproducibility design

The archive contains `fabius_multiresolution_experiments.py`, a standalone, fully commented Python program. Its arithmetic and analytic checks are intentionally separated.

- *Exact layer.* Python’s `Fraction` type evaluates (22), every $F(a/2^{16})$, all Faber coefficients below level 16, every R_J , and every coefficient of P_N for $N \leq 16$. Log-concavity is tested with integer inequalities.
- *Independent analytic layer.* The reference values of F and up come from the period-two cosine series whose coefficients are evaluated by the sinc product. Haar, triangular, cardinal, Bessel, and Kummer formulas are checked against this independent product route.
- *Outputs.* The run writes human-readable and JSON results, the TeX tables included below, and four figures. No network access or hidden data files are used.

The exact level-16 table contains 65537 rational dyadic values and is constructed by the terminating highest-bit recursion in roughly linear time in the table size. The bundled run uses 90 decimal digits for mpmath operations.

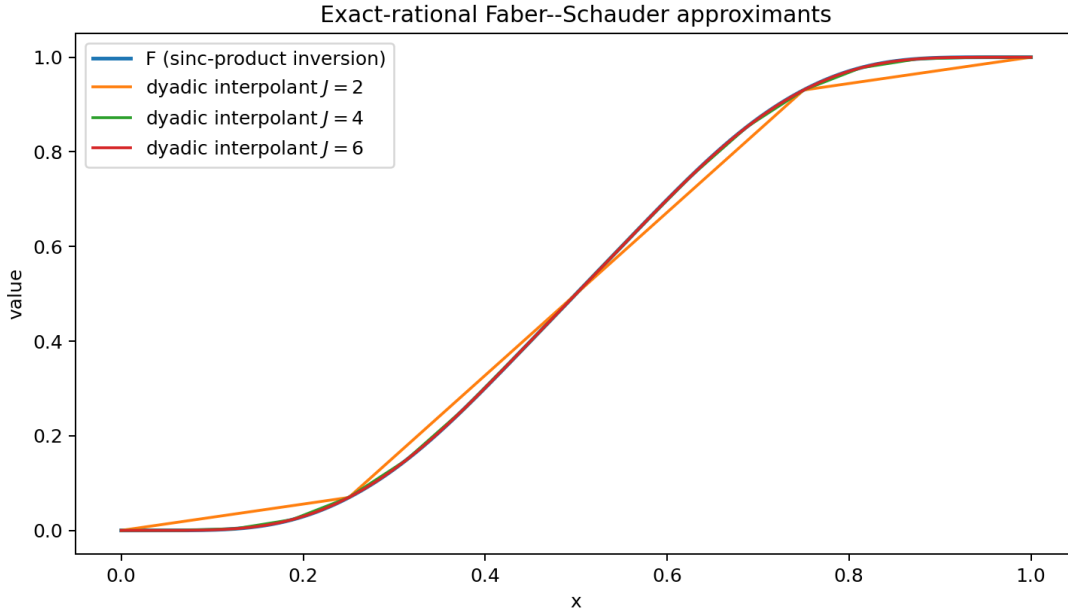


Figure 1: The exact-rational dyadic interpolants are the Faber partial sums. The level-6 curve is already visually indistinguishable from the independent sinc-product reconstruction on most of the interval.

12.2 Faber convergence

The observed maximum errors satisfy the rigorous bound 4^{-J} and the scaled errors approach the sharp constant one from (41). At level $J = 10$, the maximum grid residual against the independent Fourier evaluator is approximately 9.54×10^{-7} , consistent with $4^{-10} = 9.5367 \times 10^{-7}$.

12.3 Energy partial sums and acceleration

Table 4: Exact rational energy partial sums and accelerated values.

J	R_J (decimal)	accelerated	raw error
2	0.3804012345679012	0.414983164983165	0.024036
3	0.3960262345679012	0.4044523246650906	0.0084105
4	0.4023494663214687	0.4044383623771493	0.0020873
5	0.4039112434127894	0.4044367566681171	0.00052552
6	0.404305183886946	0.4044367679881609	0.00013158
8	0.4044285399647054	0.4044367679839853	8.228e-6
10	0.4044362537170872	0.4044367679839853	5.1427e-7
12	0.4044367358422428	0.4044367679839853	3.2142e-8
14	0.4044367659751261	0.4044367679839853	2.0089e-9
16	0.4044367678584316	0.4044367679839853	1.2555e-10

The decimal in (89) is not advertised as a formally certified 60-digit interval: the all-orders theorem is asymptotic and the bundled run does not compute explicit constants in its remainder. It is, however, stable under changes of both level and acceleration order. For example, the five-term estimates from $J = 12, 13, 14, 15, 16$ successively stabilize through roughly 8, 12, 16, 20, and 21 additional digits beyond the raw partials. A rigorous high-precision enclosure can be obtained by combining a finite

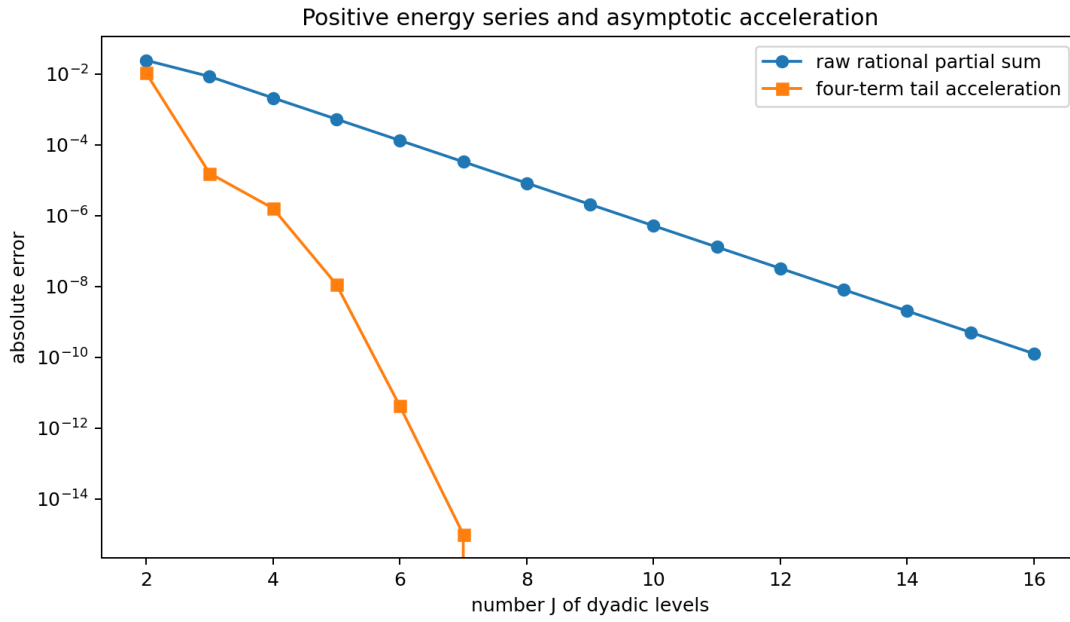


Figure 2: Raw positive rational partial sums converge geometrically. Using four terms of the all-orders tail expansion removes several additional powers of 4^{-J} . Values below machine precision relative to the numerical reference are omitted automatically from the logarithmic plot.

Table 5: Chebyshev coefficients from cosine quadrature and the Bessel–sinc integral.

n	cosine-grid value	Bessel-product value	absolute residual
0	0.68197894345	0.681978961331	1.79e-08
2	-0.514948073175	-0.514948055353	1.78e-08
4	0.195935922808	0.195935940418	1.76e-08
6	-0.00235687932482	-0.0023568621795	1.71e-08
8	-0.028210810035	-0.028210793777	1.63e-08
10	0.0114279577063	0.0114279724231	1.47e-08
12	-0.00612048206979	-0.00612046982494	1.22e-08

Table 6: Exact tests of the triangular coefficient conjecture.

N	degree	palindromic	log-concave
4	11	yes	yes
6	57	yes	yes
8	247	yes	yes
10	1013	yes	yes
12	4083	yes	yes
14	16369	yes	yes
16	65519	yes	yes

higher-order expansion with explicit derivative sup norms and an Euler–Maclaurin remainder.

12.4 Haar, triangular, and cardinal checks

Using levels $m < 14$, the Haar–sinc² formula (65) has absolute residuals between 10^{-10} and 8×10^{-9} at the tested real and complex arguments. This is the expected unresolved tail size. The finite triangular product at $N = 36$ agrees with the sinc product to the precision predicted by (98).



Figure 3: Maximum residual of the odd-sample cardinal formula at two real and one complex test point. The small nonmonotonic ripples reflect cancellation; the envelope decreases superalgebraically.

With 40 positive odd-sample pairs, the absolute residuals are 1.2×10^{-14} to 3.9×10^{-13} at the six test points listed in `numerical_results.txt`. The explicit sum rules (116) and (117) can be checked from the same sample table.

Exact integer arithmetic confirms palindromicity and log-concavity for every $2 \leq N \leq 16$; selected degrees are shown in table 6. The checks are exhaustive over every interior coefficient, not random spot tests.

12.5 Bessel and Bernstein checks

The cosine-grid and Bessel-product evaluations in table 5 agree to about 2×10^{-8} . The residual is set by the finite cosine grid and finite Bessel integration cutoff, not by a symbolic approximation in the theorem.

For the rational Bernstein approximants to F , the observed uniform errors are:

n	observed error	rigorous bound $1/n$
8	7.1353×10^{-2}	1.25×10^{-1}
16	4.1193×10^{-2}	6.25×10^{-2}
32	2.2565×10^{-2}	3.125×10^{-2}
64	1.1906×10^{-2}	1.5625×10^{-2}
128	6.1361×10^{-3}	7.8125×10^{-3}
256	3.1214×10^{-3}	3.90625×10^{-3}

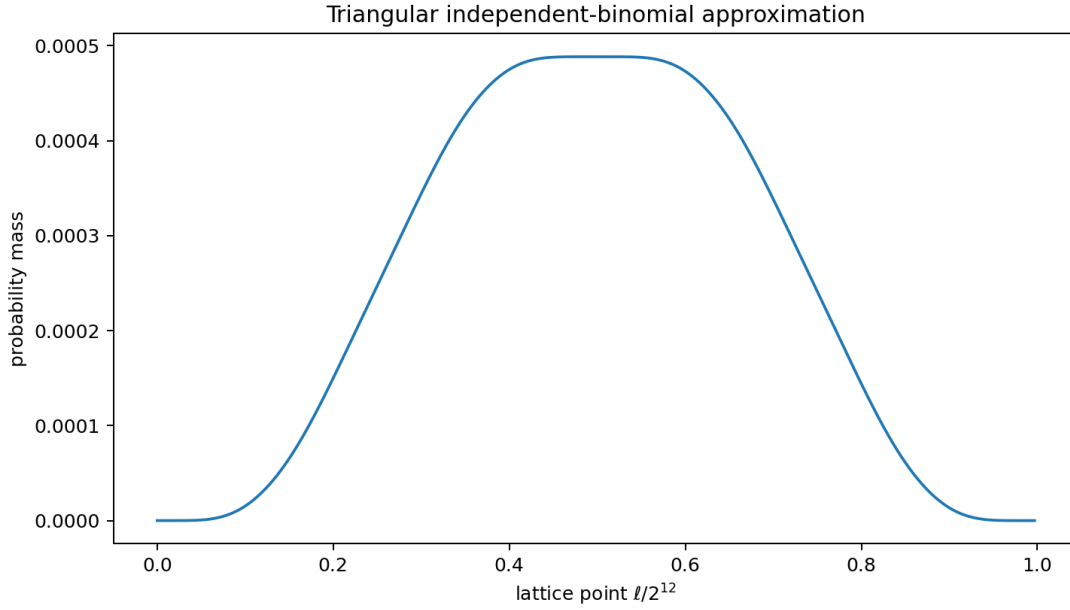


Figure 4: The positive mass distribution encoded by P_{12} . Its broad, nearly flat central region is a finite-layer effect quite different from a single binomial law.

At degree 32 and $t = 2.7$, direct quadrature of the Bernstein bump and the finite Kummer sum (136) differ by 4.5×10^{-17} .

13 Conjectures and frontier directions

13.1 Arithmetic nature of the energy constant

Conjecture 13.1 (Irrationality of the Fabius energy). *The number*

$$A_2 = \int_0^1 F(x)^2 dx$$

is irrational; plausibly it is transcendental.

The positive rational expansion (70) is more arithmetically structured than the original sinc integral. A first serious attack should study the reduced denominators of R_J , their p -adic growth, and whether the tail asymptotic can be converted into rational-approximation bounds incompatible with algebraicity. The main obstacle is that an excellent asymptotic expansion by itself does not provide the denominator separation needed for an irrationality proof.

13.2 Endpoint phase tomography through rational wavelets

For a cell entirely in $(0, 1/4)$, the differential equation $F''(x) = 8F(4x)$ and (34) give the exact scaled identity

$$-4^{m-1}c_{m,k} = \int_0^1 \min(s, 1-s) F(2^{2-m}(k+s)) ds. \quad (149)$$

Thus every fixed- k coefficient ray is a positive triangular average of the endpoint function at a dyadically shrinking scale.

Conjecture 13.2 (Wavelet recovery of the Lambert phase). *For each fixed $k \geq 0$, the logarithm of $-c_{m,k}$ has an all-orders expansion obtained by applying boundary Laplace analysis to (149). Its leading exponential part is governed by the same lower-Lambert coordinate as F , and its bounded oscillatory part is a nonconstant translate of the repository’s one-periodic endpoint correction. Distinct k determine distinct phase shifts but the same Fourier spectrum.*

This would turn exact rational coefficient rays into a tomography device for the periodic endpoint phase. It is important not to replace the phase by its mean: the audited endpoint work proves that such averaging fails at the natural error scale.

13.3 The triangular coefficient problem

Conjecture 6.4 asks for a structural proof of log-concavity of a highly nonstandard sparse product. Promising approaches include:

- induction through interlacing inequalities stronger than ordinary log-concavity;
- a probabilistic proof using ultra-log-concavity under dyadically weighted Bernoulli convolution;
- total positivity of the shift-and-add operators that multiply by $(1 + q^{2^r})^a$ when the lower scales are already present;
- a stable-polynomial or Lorentzian homogenization in extra variables, followed by a carefully justified diagonal specialization.

A local central limit theorem with explicit lattice corrections for X_N^Δ would also quantify how the positive coefficient limit approaches the smooth Fabius density.

13.4 Odd samples as intrinsic spectral coordinates

The sampling formula uses all odd samples because the even samples are forced to vanish. This suggests a constrained Paley–Wiener space

$$\mathcal{V} = \{f \in \text{PW}_1 : f(2\pi n) = 0 \text{ for } n \neq 0, \text{ with the prescribed higher 2-adic jets}\}. \quad (150)$$

Problem 13.3 (de Branges/sampling model). Construct an intrinsic reproducing-kernel or de Branges description of the closure generated by the Fabius zero divisor, and determine whether the odd samples form a natural Riesz coordinate sequence after the higher even-zero constraints are imposed. Derive sharp weighted truncation inequalities for (111) that exploit the exact zero multiplicities.

The identities in theorem 7.2 are the first algebraic constraints such a model must reproduce.

13.5 Nonlinear wavelet energies

For $p = 2$, orthogonality collapses the energy to the positive diagonal sum (70). For even $p > 2$, products of Haar functions are still finite on each dyadic tree but no longer diagonal.

Problem 13.4 (Exact dyadic tree formulas for L^p energies). Find finite-level tree recurrences and convergent rational series for

$$A_p = \int_0^1 F(x)^p dx, \quad U_p = \int_{-1}^1 \text{up}(x)^p dx,$$

starting with $p = 3, 4$. Determine whether any rearrangement is positive and whether its level tails are again controlled by the derivative-norm spectrum.

The order-statistic formulas in the current corpus evaluate many A_p -type integrals probabilistically; the wavelet problem asks for a genuinely local, multiscale arithmetic decomposition.

13.6 Orthogonal-polynomial coefficient geometry

The Chebyshev coefficients in (122) do not alternate forever in the simplest way. More nuanced questions remain.

Problem 13.5 (Sign and zero laws). Study the sign changes, asymptotic phase, and zero distribution of $a_{2r}^{(\lambda)}$ as functions of r and λ . Determine whether the Bessel integrals (126) admit saddle descriptions tied to the sharp real-axis decay rays of Φ , and identify values of λ for which the coefficients have additional arithmetic structure.

The Legendre case is especially attractive because its coefficients are rational, making (129) an exact bridge between orthogonal-polynomial arithmetic and oscillatory integral analysis.

13.7 Base- b and geometric-weight extensions

The repository already develops a base- b sinc-product and zero-spectrum theory. Every new construction above has a plausible geometric-weight analogue:

- b -adic Faber systems with exact node arithmetic;
- regrouping base- b digits along antidiagonals, producing multinomial rather than binomial layers;
- cardinal sampling after identifying the appropriate forced zero sublattice;
- Jacobi/Gegenbauer coefficients as Bessel transforms of the base- b product;
- Bernstein limits along degrees b^r .

The dyadic case is exceptional because binary digit regrouping and Haar multiresolution are perfectly aligned; identifying which conclusions survive for $b > 2$ should clarify which phenomena are genuinely dyadic.

13.8 Lean formalization roadmap

The new results are unusually formalization-friendly because their analytic core can be divided into finite exact layers and standard convergence lemmas. A practical order is:

1. Define $\Lambda_{m,k}$ and prove that the finite Faber sum is the dyadic linear interpolant. Import the existing exact dyadic evaluator to prove coefficient rationality.
2. Prove the C^2 interpolation bound and the sharp limit at $1/4$.
3. Differentiate finite interpolants, identify conditional averages of F' , and obtain L^2 convergence. Haar orthogonality then proves (70) with minimal measure-theoretic overhead.
4. Formalize the coefficient kernel (34), the uniform bound, and the first few energy-tail coefficients. The all-orders theorem can follow after an endpoint-flat Euler–Maclaurin library is in place.
5. Construct the triangular Bernoulli array on the existing infinite product probability space. Regroup nonnegative series and prove independence of antidiagonals.
6. Apply the existing Fourier product API to prove (95); keep the finite polynomial and exact coefficient checks as executable definitions.
7. Reuse a Paley–Wiener sampling theorem from mathlib if available, or first formalize the rapidly decreasing special case needed for (111). The sample-jet identities then reduce to differentiation of normally convergent rational series.
8. Add the beta–Kummer integral and Bernstein convergence as a separate approximation module; these require no endpoint asymptotics.

The energy identity is the best first target: it combines exact dyadic arithmetic, finite orthogonality, and a single L^2 limit, and it creates a new formally verified representation of a constant already central to the integral reports.

14 Conclusion

The Fabius–Rvachev system is often introduced through one striking identity: a compactly supported C^∞ function whose Fourier transform is an infinite product of sinc functions. The audited corpus already shows that this is only one face of a much larger structure involving uniform random series, Thue–Morse signs, q -binomial algebra, exact dyadic values, spectral multiplicities, Lambert– W endpoint phases, Bell polynomials, and a rich integral-transform calculus.

The new results add a multiresolution and sampling layer to that structure. Exact dyadic values are not merely isolated arithmetic curiosities: they are the coordinates of a rational Faber basis, and inversion of its finite interpolants gives exact-rational quantile splines. Their discrete derivatives are rational Haar coefficients of the smooth density. Orthogonality turns those coefficients into a positive rational expansion of the central sinc energy, and smooth self-similarity determines the complete tail expansion. Binary digits reorganize the same law into independent binomial layers; compact support reorganizes the Fourier image into odd cardinal samples; plane-wave identities reorganize it into Bessel transforms; Bernstein’s probabilistic polynomials reorganize it into finite Kummer sums; and finite-dimensional geometry reorganizes it into cube volumes and sections.

No one representation dominates all others. The sinc product is best for zeros and global frequency decay; the Faber series is best for exact local dyadic approximation; the Haar series is best for energy; the triangular layers are best for positive combinatorics; odd sampling is best for entire reconstruction from sparse spectral data; Bessel expansions connect to orthogonal approximation; Bernstein–Kummer limits provide finite entire approximants; and cube sections expose the underlying geometry. The most promising next step is to formalize the rational energy series and then use its exact denominators and all-order tail to investigate the arithmetic nature of A_2 .

A Remainder proof for the all-orders energy expansion

This appendix supplies a finite-order argument behind [theorem 5.3](#). It avoids treating the nowhere-analytic function F as if its formal differential-operator series converged.

Let

$$\Delta = 2^{-m}, \quad h = \Delta/2, \quad x_k = (k + \tfrac{1}{2})\Delta. \quad (151)$$

For an arbitrary fixed integer $M \geq 1$, Taylor’s theorem with a uniform remainder gives

$$c_{m,k} = - \sum_{r=1}^M \frac{h^{2r}}{(2r)!} F^{(2r)}(x_k) + \mathcal{O}(h^{2M+2}), \quad (152)$$

uniformly in k . Squaring this identity and retaining terms through any prescribed power of Δ is therefore legitimate.

The level energy may be written as

$$E_m = \Delta^{-2} \left[\Delta \sum_{k=0}^{\Delta^{-1}-1} c_{m,k}^2 \right]. \quad (153)$$

Every coefficient in the finite expansion of $c_{m,k}^2$ is a product $F^{(a)}F^{(b)}$ with $a, b \geq 2$. All derivatives of F of positive order vanish at 0 and 1. Hence each such product and all of its derivatives vanish at

both endpoints. The midpoint Euler–Maclaurin formula therefore has no boundary correction of any algebraic order: for every fixed smooth product G of this form and every $L \geq 1$,

$$\Delta \sum_{k=0}^{\Delta^{-1}-1} G(x_k) = \int_0^1 G(x) dx + \mathcal{O}_L(\Delta^L). \quad (154)$$

Combining (152)–(154) shows that the Poincaré coefficients of E_m are the same as those of

$$\Delta^{-2} \int_0^1 [(\cosh(hD) - 1)F(x)]^2 dx, \quad (155)$$

where the right side is interpreted only as a formal device for extracting any finite number of Taylor coefficients.

Since $\cosh(hD) - 1$ contains only even positive powers of D , repeated integration by parts has no boundary terms. Thus, coefficientwise,

$$\int_0^1 [(\cosh(hD) - 1)F]^2 dx = \int_0^1 F(\cosh(hD) - 1)^2 F dx \quad (156)$$

$$\sim \sum_{r=2}^{\infty} \frac{2^{2r-1} - 2}{(2r)!} h^{2r} (-1)^r I_r. \quad (157)$$

Now set $r = q+1$, use $h = \Delta/2$, and substitute $I_{q+1} = 2^{(q+1)(q+2)} A_2$. The coefficient of $\Delta^{2q} = 4^{-qm}$ is

$$(-1)^{q+1} \frac{2^{2q+1} - 2}{(2q+2)!} \frac{2^{(q+1)(q+2)}}{2^{2q+2}} A_2 = (-1)^{q+1} \frac{2^{q(q+1)+1}(2^{2q} - 1)}{(2q+2)!} A_2, \quad (158)$$

which is $\beta_q A_2$.

More explicitly, for every fixed $Q \geq 1$ the preceding finite-order argument gives

$$E_m = A_2 \sum_{q=1}^Q \beta_q 4^{-qm} + \mathcal{O}_Q(4^{-(Q+1)m}). \quad (159)$$

Summing the remainder over $m \geq J$ proves the finite-order version of (83) and the acceleration estimate (88).

B Exact dyadic evaluation used by the experiments

The experiments do not solve a discretized differential equation. They use the repository’s exact arithmetic in the following terminating form.

Algorithm B.1 (Highest-bit evaluator). Assume the values $V_n = F(2^{-n})$ have been computed from (22). To evaluate $F(a/2^e)$ with $0 < a < 2^e$, write

$$a = 2^b + q, \quad 0 \leq q < 2^b, \quad r = e - b, \quad y = q/2^e. \quad (160)$$

Then

$$F(a/2^e) = \sum_{j=0}^r \frac{2^{j(j+1)/2} V_{r-j}}{j!} y^j - F(q/2^e). \quad (161)$$

Because $q < a$, a left-to-right table of all numerators memoizes every subproblem and terminates after clearing the set bits of a .

For a fixed exponent e , the algorithm computes all $2^e + 1$ values using $\mathcal{O}(e2^e)$ rational operations. A single level-16 table then supplies all coarser Faber coefficients by exact subsampling.

C Compact formula compendium

For convenience, the principal formulas are collected here without proofs. The status column refers to the pinned corpus boundary, not to absolute historical priority.

Table 7: Formula compendium.

Object	Formula	Status
F	$F(x) = \mathbb{P}(\sum_{j \geq 1} 2^{-j} U_j \leq x)$	baseline
up	$\text{up}(y) = *_{j \geq 1} 2^{j-1} \mathbf{1}_{[-2^{-j}, 2^{-j}]}(y)$	baseline
Φ	$\Phi(t) = \prod_{j \geq 1} \text{sinc}(t/2^j)$	baseline
Φ	$\Phi(t) = \prod_{n \geq 1} (1 - t^2/(4\pi^2 n^2))^{1+\nu_2(n)}$	baseline
F_N	$\frac{2^{N(N+1)/2}}{N!} \sum_{k < 2^N} (-1)^{\text{wt}_2(k)} (x - k/2^N)_+^N$	baseline
F	$x + \sum_{m,k} c_{m,k} \Lambda_{m,k}(x)$	new
G	$\lim_J [s_{J,0} y + \sum_k (s_{J,k} - s_{J,k-1})(y - y_{J,k})_+]$	new
up	$1 - y + \sum_{m,k} c_{m,k} \Lambda_{m,k}(1 - y)$	new
F'	$1 + \sum_{m,k} 2^{m+1} c_{m,k} H_{m,k}$	new
Φ	$\text{sinc } t + \sum_{m,k} \frac{2^m c_{m,k}}{it} e^{it(1-2k/2^m)} (1 - e^{-it/2^m})^2$	new
A_2	$\frac{1}{4} + \sum_{m \geq 0} 2^m \sum_k c_{m,k}^2$	new
E_m	$E_m \sim A_2 \sum_{q \geq 1} \beta_q 4^{-qm}$	new
X	$\sum_{n \geq 2} 2^{-n} K_n, K_n \sim \text{Bin}(n-1, 1/2)$ independent	new
Φ	$\prod_{n \geq 2} \cos(t/2^n)^{n-1}$	new
F	$\lim_N 2^{-N(N-1)/2} \sum_{\ell \leq 2^N x} [q^\ell] P_N(q)$	new
Φ	$\frac{\sin z}{z} - 2z \sin z \sum_{m \geq 0} \frac{s_m}{z^2 - (2m+1)^2 \pi^2}$	new
a_{2r}	$\frac{2(-1)^r}{\pi} \int_0^\infty J_{2r}(t) \Phi(t) dt$	new
$A(t)$	$\lim_{r \rightarrow \infty} \frac{1}{2^r + 1} \sum_k F(k/2^r) {}_1F_1(k+1; 2^r+2; -it)$	new
Φ	Dyadic Bernstein–Kummer limit (137)	new
F	Weighted-cube sublevel-volume limit (139)	new packaging
up	Weighted-cube hyperplane-section limit (144)	new packaging

D Corpus and reproducibility files

The accompanying archive contains:

`Fabius_Rvachev_Multiresolution_Representations.tex`

The complete source of this report.

`Fabius_Rvachev_Multiresolution_Representations.pdf`

The compiled report.

`fabius_multiresolution_experiments.py`

Exact and numerical verification code with detailed comments.

`numerical_results.txt` and `numerical_results.json`

Human-readable and machine-readable output of the bundled run.

`generated_tables.tex`

The tables included by [section 12](#).

`figures/`

Four plots generated by the script.

`corpus_manifest_current.txt`

The 92-path recursive TeX manifest, the five additions after the baseline snapshot, and the modified-path comparison ledger.

`README.md`, `requirements.txt`, and `SHA256SUMS.txt`

Reproduction instructions, pinned Python dependencies, and checksums.

The command used for the bundled numerical run is

```
python fabius_multiresolution_experiments.py \
    --max-exact-level 16 --precision 90
```

The program was executed with Python 3, NumPy, SciPy, mpmath, and Matplotlib. It does not access the network.

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